

ASSIGNMENT-5

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① $n=20,$

$$\sum x = 51.5$$

$$\sum y = 298.24$$

$$\sum x^2 = 190.15$$

$$\sum y^2 = 5489.464$$

$$\sum xy = 1008.628.$$

$$\bar{x} = \frac{\sum x}{n} = \frac{51.5}{20} = 2.575$$

$$\bar{y} = \frac{\sum y}{n} = \frac{298.24}{20} = 14.912$$

② $S_{xx} = \sum x^2 - n(\bar{x})^2 = 190.15 - 20(2.575)^2 = 57.5375$

$$S_{yy} = \sum y^2 - n(\bar{y})^2 = 5489.464 - 20(14.912)^2 = 1042.1091$$

$$S_{ny} = \sum xy - n(\bar{x}\bar{y}) = 1008.628 - 20(2.575)(14.912) = 240.66$$

$$m = \frac{S_{ny}}{S_{xx}} = \frac{240.66}{57.5375} = 4.1827$$

$$c = \bar{y} - m\bar{x} = 14.912 - (4.1827)(2.575) = 4.1416$$

$$SSE = S_{yy} - mS_{ny} = 1042.1091 - (4.1827)(240.66) = 35.509.$$

$$s^2 = \frac{SSE}{n-2} = \frac{35.509}{18} = 1.9727$$

⑥ If we observe the residual plot like it shows a random scatter of points around the zero line with no clear curved patterns.

↳ so, we can say that residuals are independent and have constant variance.

⑦ The histogram of the residuals is roughly symmetric and they are centred about zero.

↳ This implies error terms follow a normal distribution.

$$\textcircled{8} R^2 = \frac{S_{yy} - SSE}{S_{yy}}$$

$$R^2 = \frac{1042 \cdot 1091 - 35.509}{1042 \cdot 1091} = \frac{1006.6001}{1042 \cdot 1091} = 0.9659$$

$$\textcircled{9} \hat{y} = 4.1827(7) + 4.1416 = 33.4205$$

$$SE = \sqrt{s^2 \left(\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right)}$$

$$SE = \sqrt{1.9727 \left(\frac{1}{20} + \frac{(7 - 2.575)^2}{57.5375} \right)}$$

$$SE = \sqrt{1.9727(0.05 + 0.3403)} = 0.8777$$

$$df = 20 - 2 = 18$$

$$t_{0.05, 18} = 1.734$$

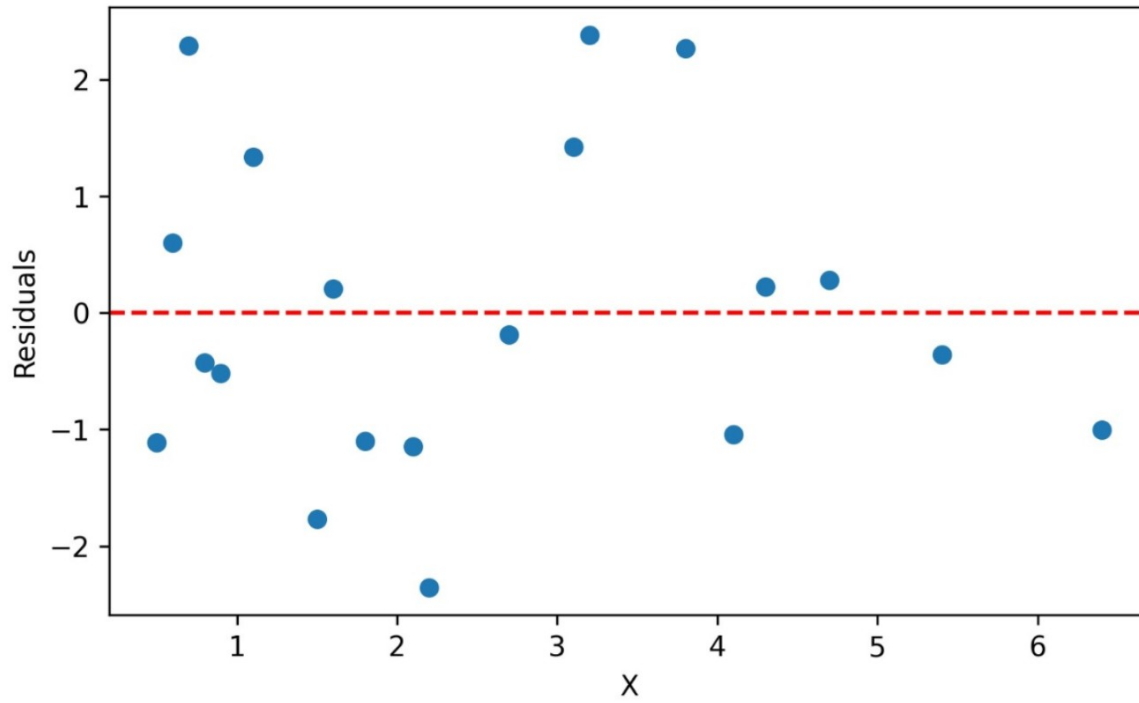
$$ME = 1.734 \times 0.8777 = 1.5214$$

$$LB = 33.4205 - 1.5214 = 31.8991$$

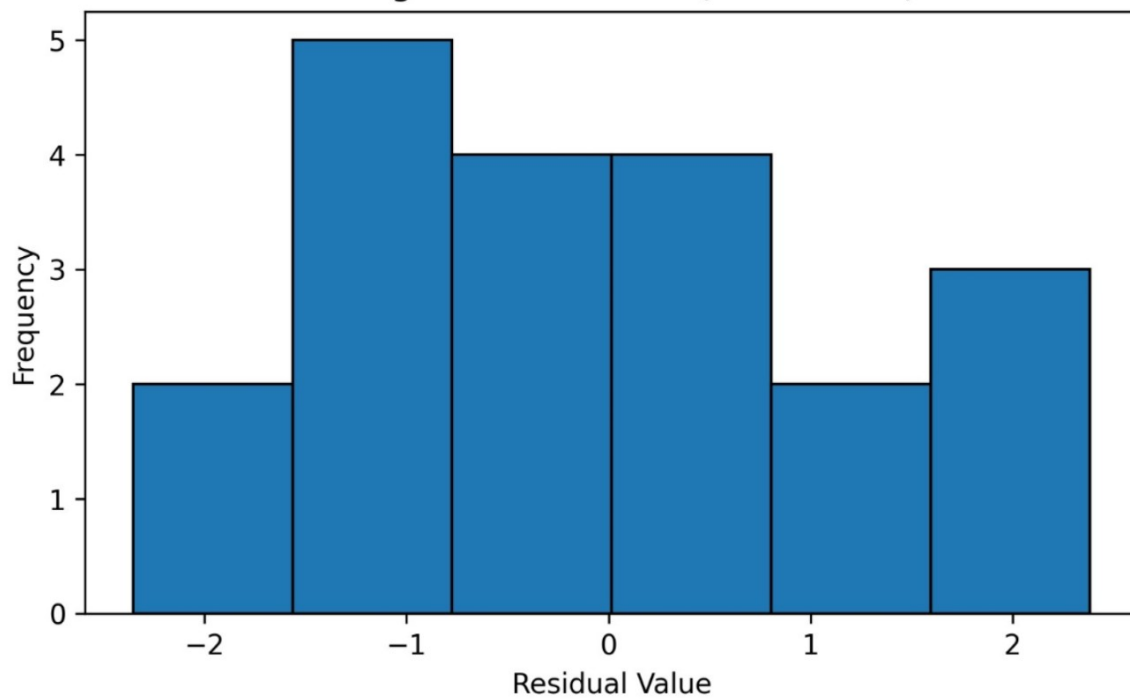
$$UB = 33.4205 + 1.5214 = 34.9419$$

$$\text{Interval} = [31.8991, 34.9419]$$

Residuals vs X (Independence)



Histogram of Residuals (Distribution)



$$\textcircled{2} \quad y = ax^b$$

$$\ln y = \ln a + b \ln x$$

$$Y = \ln y$$

$$Y = MX + C$$

$$X = \ln x$$

$$M = b$$

$$C = \ln(a)$$

x_{raw}	y_{raw}	X	Y	X^2	XY
0.1	2	-2.3026	0.6931	5.3020	-1.5960
0.2	2.47	-1.6094	0.9042	2.5902	-1.4552
0.4	3.04	-0.9163	1.1119	0.8396	-1.0188
0.6	3.43	-0.5108	1.2326	0.2609	-0.6296
0.9	3.88	-0.1054	1.3558	0.0111	-0.1424
1.3	4.33	0.2624	1.4656	0.0689	0.2846
1.5	4.52	0.4055	1.5085	0.1644	0.6117
1.7	4.69	0.5306	1.5454	0.2815	0.8200
1.8	4.77	0.5878	1.5623	0.3455	0.9183
$n = 9$		$\sum X = -2.6582$	$\sum Y = 11.2794$	$\sum X^2 = 9.8641$	$\sum XY = -2.1079$

$$\bar{x} = \frac{-1.6582}{9} = -0.4065$$

$$\bar{y} = \frac{11.3744}{9} = 1.2644$$

$$\begin{aligned} S_{xx} &= \sum x^2 - n(\bar{x})^2 = 9.8641 - 9(-0.4065)^2 \\ &= 9.8641 - 1.4871 = 8.3770 \end{aligned}$$

$$\begin{aligned} S_{xy} &= \sum xy - n(\bar{x}\bar{y}) = -2.1079 - 9(-0.4065)(1.2644) \\ &= -2.1079 - (-4.6257) = 2.5178 \end{aligned}$$

$$M = \frac{S_{xy}}{S_{xx}} = \frac{2.5178}{8.3770} = 0.3006$$

$$\begin{aligned} C = \bar{y} - M\bar{x} &= 1.2644 - (0.3006)(-0.4065) \\ &= 1.2644 + 0.1222 = 1.3866 \end{aligned}$$

$$a = b = 0.3006$$

$$A = e^C \Rightarrow A = e^{1.3866} = 4.0012$$

$$\hat{\cdot} = \boxed{a = 4.0012 \quad b = 0.3006}$$